This is CS50.

7-SQL

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Week 7 : SQL

Outline :

- Relational Databases
- SQL
- Indexes
- Using SQL in Python

Relational Databases

سهلة و سريعة بس غير منظمة و مش بتعرف تتعامل مع الداتا الكبيرة

flat-file database	Relational Databases
data is stored in individual files or records.	data is organized into tables with rows and columns.
not designed to handle complex data management tasks.	more powerful and flexible, providing more advanced data management capabilities.
less scalable, can't handle large amounts of data.	more scalable, can handle large amounts of data.
many copies, duplicating and losing data.	update one copy , avoid duplicating and losing data.

اكثر تعقيدا بس قوية و مرنة اكثر فى التعامل مع الداتا الكبيرة

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Relational Databases

النوع بقا الى هنشتغل عليه من انواع الdatabase هى ال relational database و ده النوع الى بتستخدمه شركة جوجل و تويتر و ميتا فى تخزين بياناتهم بشكل اكبر

- Relational databases store data in rows and columns in structures called tables.
- A database most often contains one or more tables. Each table is identified by a name, columns, and contain records (rows) with data.

Relational Databases

- A record, also called a row, is each individual entry that exists in a table. A record is a horizontal entity in a table.

#########	Python	Hello, World

- ال record او ال raw يمثل صف فى الجدول

- A column is a vertical entity in a table that contains all information associated with a specific field in a table.

language
Python
Python
Python
Scratch
Python
Python
Python
Python
Python
Python

- ال column يمثل عمود فى الجدول

SQL

What is SQL?

- Most of the actions you need to perform on a database are done with SQL statements.
- SQL stands for Structured Query Language
- SQL lets you access and manipulate databases

Most Important SQL Commands

- CREATE : create a new table in a database.
- READ : read data from a database.
- UPDATE : updates data in a database.
- DELETE : deletes data from a database.
- SELECT : return data from a database.

- و الاربع commands دول تحديدا ليهم مسمى و هو CRUD

SQL

- In sqlite, we have five datatypes, including:

```
BLOB        -- binary large objects that are groups of ones and zeros
INTEGER     -- an integer
NUMERIC     -- for numbers that are formatted specially like dates
REAL        -- like a float
TEXT        -- for strings and the like
```

- SQL offers additional commands we can utilize in our queries:

```
WHERE       -- adding a Boolean expression to filter our data
LIKE        -- filtering responses more loosely
ORDER BY    -- ordering responses
LIMIT       -- limiting the number of responses
GROUP BY    -- grouping responses together
```

- Notice that we use -- to write a comment in SQL.

CREATE

- The CREATE DATABASE statement is used to create a new SQL database.

```
CREATE DATABASE databasename;
```

اول command هنتعلمه و هو create
ازاى اصلا ننشئ database او قاعدة بيانات
جديدة او ننشئ table او جدول جديد

Example

```
CREATE DATABASE testDB;
```

- The CREATE TABLE statement is used to create a new table in a database.

```
CREATE TABLE table_name (
    column1 datatype,
    column2 datatype,
    column3 datatype,
);
```

Example

```
CREATE TABLE Persons (
    PersonID int,
    LastName text,
    FirstName text,
);
```

PersonID	LastName	FirstName

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PRIMARY KEY

- The PRIMARY KEY constraint uniquely identifies each record in a table.
- Primary keys must contain UNIQUE values, and cannot contain NULL values.
- A table can have only ONE primary key; and in the table, this primary key can consist of single or multiple columns (fields).

```
• CREATE TABLE Persons (
•    ID int NOT NULL,
•    Age int,
•     PRIMARY KEY (ID)
• );
```

ال primary key بيبقا زى المعرف او
القيمة المميزة الى بتميز كل جدول و
كل جدول بيبقا ليه primary key
واحد بس

Relational Databases

FOREIGN KEY

- A FOREIGN KEY is a field (or collection of fields) in one table, that refers to the <u>PRIMARY KEY</u> in another table.
- The table with the foreign key is called the child table, and the table with the primary key is called the referenced or parent table.

ال foreign key بيعرفنا ان ال
column ده كان primary key فى
جدول تانى

```
• CREATE TABLE Orders (
•    OrderID int NOT NULL,
•    OrderNumber int NOT NULL,
•    PersonID int,
•     PRIMARY KEY (OrderID),
•     FOREIGN KEY (PersonID)
  REFERENCES Persons(PersonID)
• );
```

- و لغة SQL بتتيح لينا التعامل مع النوع ده من ال database

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SELECT

- The SELECT statement is used to select data from a database.

```
• SELECT column1, column2, ... FROM table_name;
```

CustomerName
Alfreds Futterkiste
Ana Trujillo Emparedados y helados
Antonio Moreno Taquería
Around the Horn
Berglunds snabbköp

نيجى بقا عند ال command الى اسمه select و
ده اكتر command هنستعمله و ده باختصار
بيعرضلى الداتا الى انا عايزها من الداتا بيز بتاعتى

Example

```
• SELECT CustomerName FROM Customers;
```

- The SELECT * statement is used to return all columns.

```
• SELECT * FROM table_name;
```

Example

```
• SELECT * FROM Customers;
```

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
1	Alfreds Futterkiste	Maria Anders	Obere Str. 57	Berlin	12209	Germany
2	Ana Trujillo Emparedados y helados	Ana Trujillo	Avda. de la Constitución 2222	México D.F.	05021	Mexico
3	Antonio Moreno Taquería	Antonio Moreno	Mataderos 2312	México D.F.	05023	Mexico
4	Around the Horn	Thomas Hardy	120 Hanover Sq.	London	WA1 1DP	UK
5	Berglunds snabbköp	Christina Berglund	Berguvsvägen 8	Luleå	S-958 22	Sweden

WHERE

- The WHERE clause is used to filter records.

```
SELECT column1, column2, ...
FROM table_name
WHERE condition;
```

- ال where هنا بنستخدمها عشان نحدد البيانات الى احنا عايزينها بالظبط

Example

```
SELECT * FROM Customers
WHERE Country='Mexico';
```

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
2	Ana Trujillo Emparedados y helados	Ana Trujillo	Avda. de la Constitución 2222	México D.F.	05021	Mexico
3	Antonio Moreno Taquería	Antonio Moreno	Mataderos 2312	México D.F.	05023	Mexico
13	Centro comercial Moctezuma	Francisco Chang	Sierras de Granada 9993	México D.F.	05022	Mexico
58	Pericles Comidas clásicas	Guillermo Fernández	Calle Dr. Jorge Cash 321	México D.F.	05033	Mexico
80	Tortuga Restaurante	Miguel Angel Paolino	Avda. Azteca 123	México D.F.	05033	Mexico

LIMIT

- The LIMIT keyword is used to limit the number of data .

```
SELECT column1,
FROM table_name
LIMIT number;
```

- و هنا مثلا لو عايزين نستخرج عدد معين من الداتا بنستخدم limit

Example

```
SELECT * FROM Customers
LIMIT 3;
```

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
1	Alfreds Futterkiste	Maria Anders	Obere Str. 57	Berlin	12209	Germany
2	Ana Trujillo Emparedados y helados	Ana Trujillo	Avda. de la Constitución 2222	México D.F.	05021	Mexico
3	Antonio Moreno Taquería	Antonio Moreno	Mataderos 2312	México D.F.	05023	Mexico

LIKE

- The LIKE operator is used in a WHERE clause to search for a specified pattern in a column.

There are two wildcards often used in conjunction with the LIKE operator:

- The percent sign % represents zero, one, or multiple characters
- The underscore sign _ represents one, single character

- ال like بنستخدمها مع where عشان ندور على داتا معينة

```
SELECT column1, column2, ...
FROM table_name
WHERE columnN LIKE pattern;
```

Example

```
SELECT * FROM Customers
WHERE CustomerName LIKE 'a%';
```

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
1	Alfreds Futterkiste	Maria Anders	Obere Str. 57	Berlin	12209	Germany
2	Ana Trujillo Emparedados y helados	Ana Trujillo	Avda. de la Constitución 2222	México D.F.	05021	Mexico
3	Antonio Moreno Taquería	Antonio Moreno	Mataderos 2312	México D.F.	05023	Mexico
4	Around the Horn	Thomas Hardy	120 Hanover Sq.	London	WA1 1DP	UK

- زى هنا مثلا احنا بنبحث فى جدول ال customers عن الأسماء الى بتبدأ بحرف ال a

DISTINCT

- The SELECT DISTINCT statement is used to return only distinct (different) values.

```
SELECT DISTINCT column1,
FROM table_name;
```

- بترجعلنا القيم المختلفة بس من الجدول مش كل القيم

Example

```
SELECT DISTINCT Country
FROM Customers;
```

Country
Argentina
Austria
Belgium
Brazil
Canada
Denmark
Finland
France

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GROUP BY

- The GROUP BY statement groups rows that have the same values into summary rows, like "find the number of customers in each country".
- The GROUP BY statement is often used with aggregate functions (COUNT(), MAX(), MIN(), SUM(), AVG()) to group the result-set by one or more columns.

```
SELECT column_name(s)
FROM table_name
WHERE condition
GROUP BY column_name(s);
```

- لو عايزين مثلا نجمع القيم الى شبه بعض فى صفوف مثلا زى ان احنا نعرف عدد العملاء الى عندنا فى كل بلد بنستعمل command اسمه Group by

Example

```
SELECT COUNT(CustomerID), Country
FROM Customers
GROUP BY Country;
```

Expr1000	Country
3	Argentina
2	Austria
2	Belgium
9	Brazil
3	Canada
2	Denmark
2	Finland
11	France
11	Germany

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ORDER BY

- The ORDER BY keyword is used to sort the result-set in ascending or descending order.

```
SELECT column1, column2, ...
FROM table_name
ORDER BY column1, column2, ... ASC|DESC;
```

- و دى بتستخدم عشان نرتب الداتا بتاعتنا سواء تصاعدى asc او تنازلى desc

Example

```
SELECT * FROM Products
ORDER BY Price ASC;
```

ProductID	ProductName	SupplierID	CategoryID	Unit	Price
33	Geitost	15	4	500 g	2.5
24	Guaraná Fantástica	10	1	12 - 355 ml cans	4.5
13	Konbu	6	8	2 kg box	6
52	Filo Mix	24	5	16 - 2 kg boxes	7
54	Tourtière	25	6	16 pies	7.45
75	Rhönbräu Klosterbier	12	1	24 - 0.5 l bottles	7.75
23	Tunnbröd	9	5	12 - 250 g pkgs.	9
19	Teatime Chocolate Biscuits	8	3	10 boxes x 12 pieces	9.2

```
SELECT * FROM Products
ORDER BY Price DESC;
```

ProductID	ProductName	SupplierID	CategoryID	Unit	Price
38	Côte de Blaye	18	1	12 - 75 cl bottles	263.5
29	Thüringer Rostbratwurst	12	6	50 bags x 30 sausgs.	123.79
9	Mishi Kobe Niku	4	6	18 - 500 g pkgs.	97
20	Sir Rodney's Marmalade	8	3	30 gift boxes	81
18	Carnarvon Tigers	7	8	16 kg pkg.	62.5
59	Raclette Courdavault	28	4	5 kg pkg.	55
51	Manjimup Dried Apples	24	7	50 - 300 g pkgs.	53
62	Tarte au sucre	29	3	48 pies	49.3

AVG

- The AVG() function returns the average value of a numeric column.

```
SELECT AVG(column_name)
FROM table_name
WHERE condition;
```

- هنا لو عايزين نجيب متوسط القيم بتاعتنا فى جدول فيه قيم عددية

Example

```
SELECT AVG(Price)
FROM Products;
```

Expr1000
28.8664

COUNT

- The COUNT() function returns the number of rows that matches a specified criterion.

```
SELECT COUNT(column_name)
FROM table_name
WHERE condition;
```

- بنستخدم count عشان نعرف عدد الداتا الموجودة عندنا فى عمود معين مثلا

Example

```
SELECT COUNT(*)
FROM Products;
```

Expr1000
77

MAX

- The MAX() function returns the largest value of the selected column.

```
SELECT MAX(column_name)
FROM table_name
WHERE condition;
```

- بنستخدم ال max لو عايزين نجيب اكبر قيمة موجودة فى العمود بتاعنا

Example

```
SELECT MAX(Price)
FROM Products;
```

Expr1000
263.5

MIN

- The MIN() function returns the smallest value of the selected column.

```
SELECT MIN(column_name)
FROM table_name
WHERE condition;
```

- بنستخدم ال min لو عايزين نجيب اصغر قيمة موجودة فى العمود بتاعنا

Example

```
SELECT MIN(Price)
FROM Products;
```

Expr1000
2.5

UPDATE

- The UPDATE statement is used to modify the existing records in a table.

```
UPDATE table_name
SET column1 = value1, column2 =
value2, ...
WHERE condition;
```

- هنا لو احنا عايزين نعدل الداتا فى الجدول بتاعنا بنستخدم update

Example

```
UPDATE Customers
SET ContactName = 'Alfred Schmidt',
City= 'Frankfurt'
WHERE CustomerID = 1;
```

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
1	Alfreds Futterkiste	Alfred Schmidt	Obere Str. 57	Frankfurt	12209	Germany
2	Ana Trujillo Emparedados y helados	Ana Trujillo	Avda. de la Constitución 2222	México D.F.	05021	Mexico
3	Antonio Moreno Taquería	Antonio Moreno	Mataderos 2312	México D.F.	05023	Mexico

DELETE

- The DELETE statement is used to delete existing records in a table.

```
DELETE FROM table_name WHERE condition;
```

- هنا ال delete لو عايزين نحذف داتا موجودة عندنا

Example

```
DELETE FROM Customers WHERE CustomerName='Alfreds Futterkiste';
```

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
1	Alfreds Futterkiste	Alfred Schmidt	Obere Str. 57	Frankfurt	12209	Germany
2	Ana Trujillo Emparedados y helados	Ana Trujillo	Avda. de la Constitución 2222	México D.F.	05021	Mexico
3	Antonio Moreno Taquería	Antonio Moreno	Mataderos 2312	México D.F.	05023	Mexico

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
2	Ana Trujillo Emparedados y helados	Ana Trujillo	Avda. de la Constitución 2222	México D.F.	05021	Mexico
3	Antonio Moreno Taquería	Antonio Moreno	Mataderos 2312	México D.F.	05023	Mexico

JOINs

- A JOIN clause is used to combine rows from two or more tables, based on a related column between them.
- Let's look at a selection from the "Orders" table:

- لو بصينا على الجدولين الى قدامنا هنلاقى ان فى عمود مشترك ما بينهم و هو customerID و ده خلا ان بقا فى علاقة ما بين الجدولين دول

OrderID	CustomerID	OrderDate
10308	2	1996-09-18
10309	37	1996-09-19
10310	77	1996-09-20

Then, look at a selection from the "Customers" table:

CustomerID	CustomerName	ContactName	Country
1	Alfreds Futterkiste	Maria Anders	Germany
2	Ana Trujillo Emparedados y helados	Ana Trujillo	Mexico
3	Antonio Moreno Taquería	Antonio Moreno	Mexico

JOINs

- The JOIN clause is used to selects records that have matching values in both tables:

```
SELECT column_name, column_name, column_name
FROM table2_name
JOIN table1_name ON table2_name.column_name=table1_name.column_name;
```

- ال join بتعرضلنا القيم المشتركة ما بين جدولين عن طريق انا بنحدد العمود المشترك ما بين الجدولين دول

Example

```
SELECT Orders.OrderID, Customers.CustomerName, Orders.OrderDate
FROM Orders
JOIN Customers ON Orders.CustomerID=Customers.CustomerID;
```

OrderID	CustomerName	OrderDate
10308	Ana Trujillo Emparedados y helados	9/18/1996
10365	Antonio Moreno Taquería	11/27/1996
10383	Around the Horn	12/16/1996
10355	Around the Horn	11/15/1996
10278	Berglunds snabbköp	8/12/1996
10280	Berglunds snabbköp	8/14/1996
10384	Berglunds snabbköp	12/16/1996
10436	Blondel père et fils	2/5/1997

Indexes

- Indexes are used to retrieve data from the database more quickly than otherwise. The users cannot see the indexes, they are just used to speed up searches/queries.

```
CREATE INDEX index_name
ON table_name (column1, column2, ...);
```

على الرغم من اننا فى ال relational database قدرنا نتحكم بشكل اقوى و اسرع فى البيانات من ال CSV الا اننا ممكن نحسن البيانات فى الجداول بشكل اكبر كمان باستخدام حاجة اسمها indexes
و ال indexes بشكل اساسى بتسرع ال queries بتاعتنا

Example

```
CREATE INDEX idx_lastname
ON Persons (LastName);
```

Sqlite3

- و دى النسخة الى هنستعملها فى الكورس عشان نطبق عليها ال commands بتاعت لغة ال SQL.
- عشان نشغل برنامج ال sqlite3 بنحتاج نكتب عندنا فى ال terminal :

Sqlite3 database_name.db

Example

Sqlite3 favourites.db

- Upon being prompted, we will agree that we want to create favorites.db by pressing y.
- You will notice a different prompt as we are now inside a program called sqlite3.

Sqlite3

طب لو احنا عندنا ملف csv و عايز احوله ل database اعمل اى ؟

- هنحول برنامج ال sqlite3 لمود ال csv عن طريق ان احنا نكتب : .mode csv
- بعد كده هنعمل import لملف ال csv بتاعنا الى عايزين نحوله ل database

.import favourites.csv favourites

- It seems that nothing has happened!
- type .schema to see the structure of the database.

ال command ده بيخيلنا نعرف البنية بتاعت الداتا بيز بتاعتنا و الجداول و الأعمدة الى بتتكون منها

Using SQL in Python

- db = SQL("sqlite:///favorites.db") provide Python the location of the database file. Then, the line that begins with rows executes SQL commands utilizing db.execute.

- هنا بنستخدم ال function db.execute عشان نعمل execute لل commands بتاعت ال sql

```python
# Searches database popularity of a problem

import csv

from cs50 import SQL

# Open database
db = SQL("sqlite:///favorites.db")

# Prompt user for favorite
favorite = input("Favorite: ")

# Search for title
rows = db.execute("SELECT COUNT(*) AS n FROM favorites WHERE problem LIKE ?", favorite)

# Get first (and only) row
row = rows[0]

# Print popularity
print(row["n"])
```

THANK YOU!

BY :

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